Qualia Motion in Fourier Space: Formalising Linear, Nondirected and Contrapuntal Ambiguity in Schoenberg's Op. 19, No. 1

MUSIC AS INTERDISCIPLINE | UC COLLOQUIUM

24.09.25

Samuel Pereira

Gilberto Bernardes

José Oliveira Martins

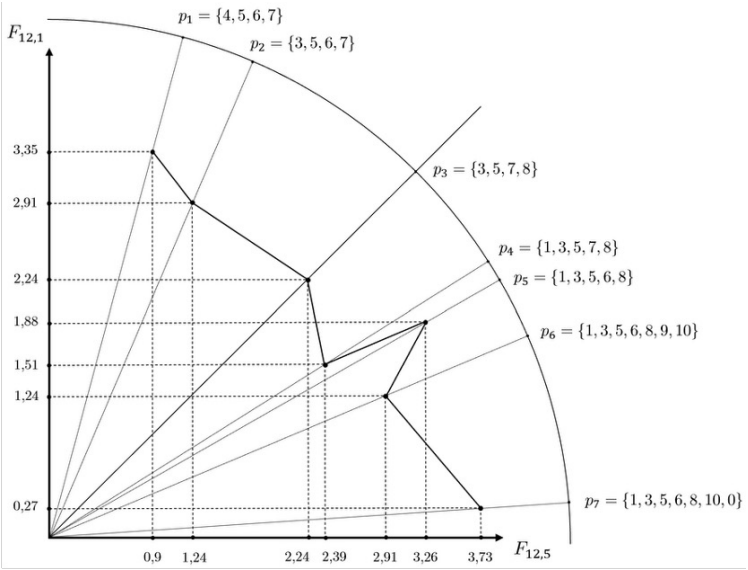
U.PORTO

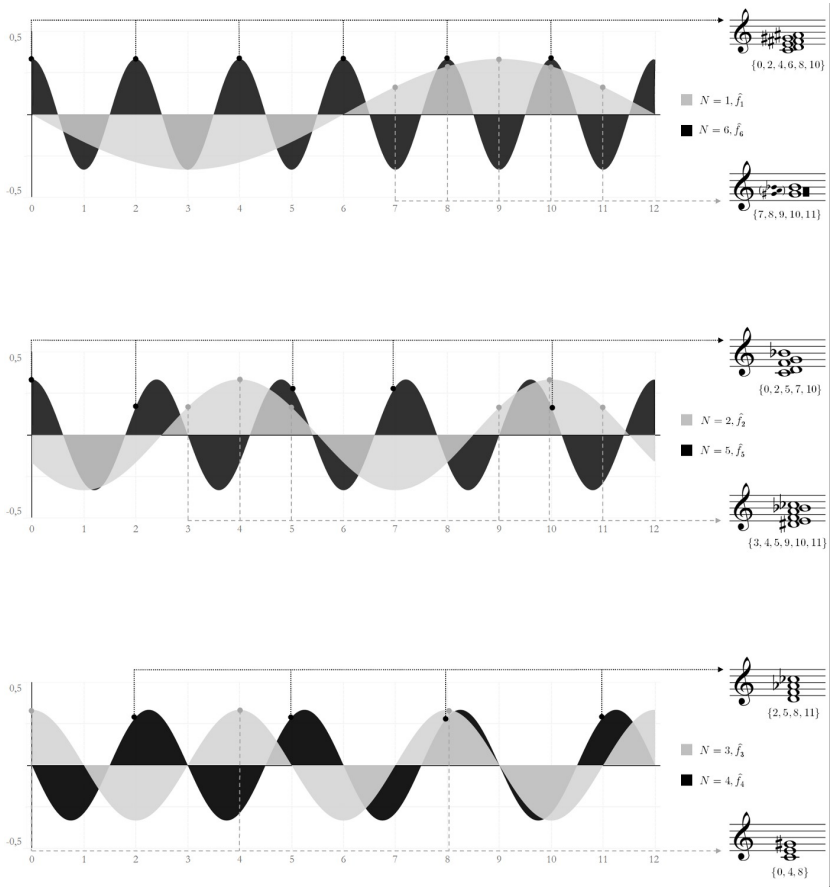
1 2 9 0 UNIVERSIDADE D COIMBRA

FCT Fundação para a Ciência e a Tecnologia

The Challenge: Conceptualise Qualia Motion

- How do we track qualia motion, especially on atonal music?
- Quinn's *quality space* is limited to only two DFT harmonic qualities at once.
- Need: Simultaneous interaction of all 6 DFT harmonic qualities.
- Goal: Track qualia motion and provide a measurable and objective analysis of qualia ambiguity.



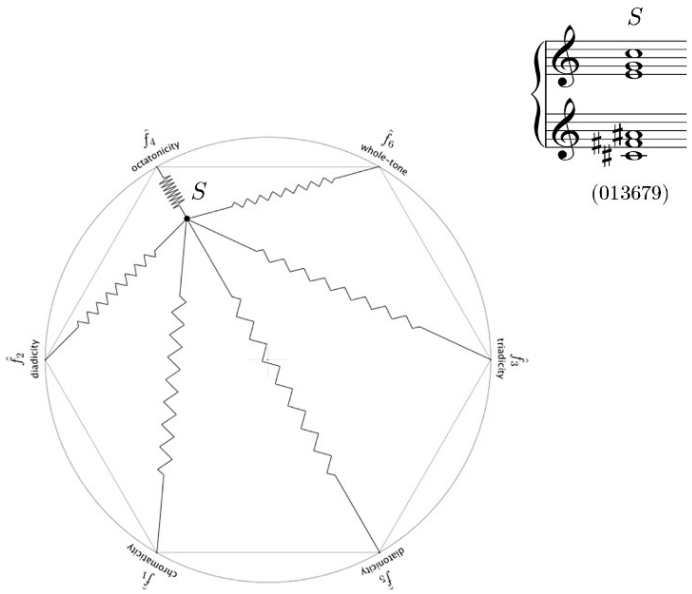


The DFT of Pitch-Class Sets

- DFT Coefficient 1 – Chromaticity.
- DFT Coefficient 2 – Dyadicity/Quartal.
- DFT Coefficient 3 – Triadicity/Hexatonicity.
- DFT Coefficient 4 – Octatonicity/Diminished.
- DFT Coefficient 5 – Diatonicity/Pentatonicity.
- DFT Coefficient 6 – Whole-tone.

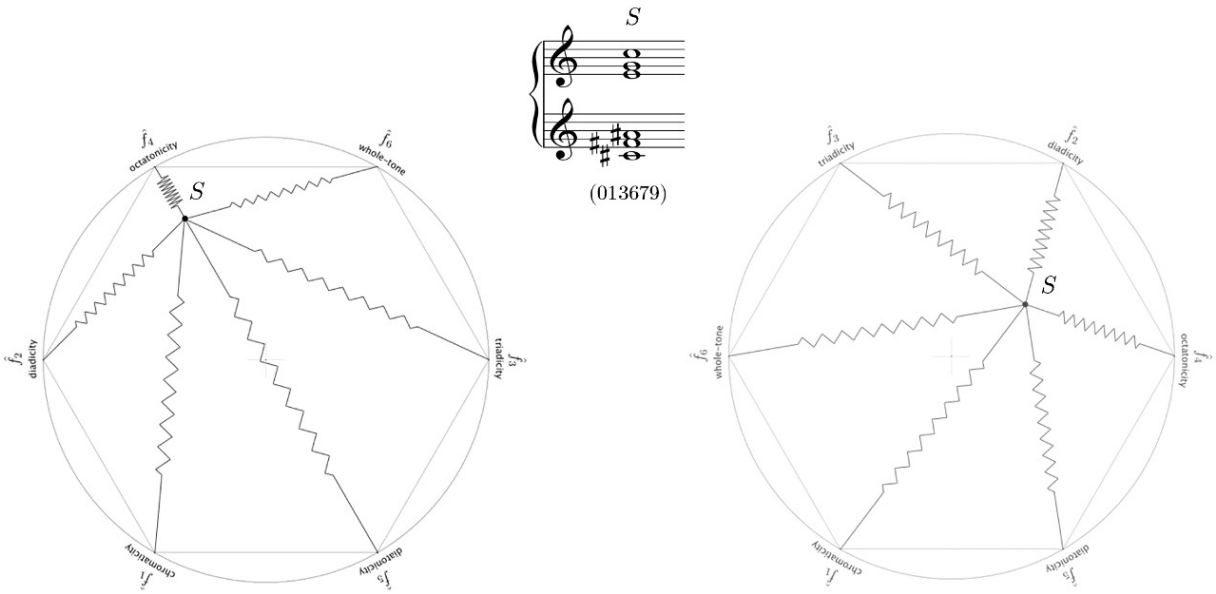
Contribution: The Fourier Qualia Space (FQS)

- *RadViz* technique: 6D $\mapsto$ 2D projection.
- Spring model with DFT magnitudes as forces.

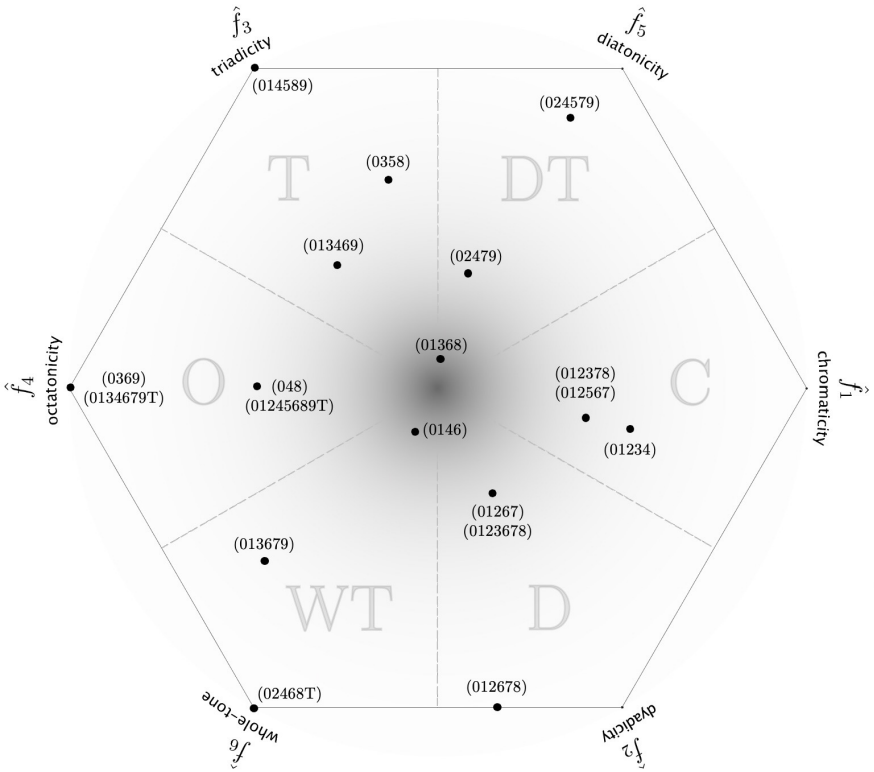


Contribution: The Fourier Qualia Space (FQS)

- *RadViz* technique: 6D $\mapsto$ 2D projection.
- Spring model with DFT magnitudes as forces.
- Transposition-independent (set class space).
- Pearson correlation coefficient $r = .729$ with 6D space.

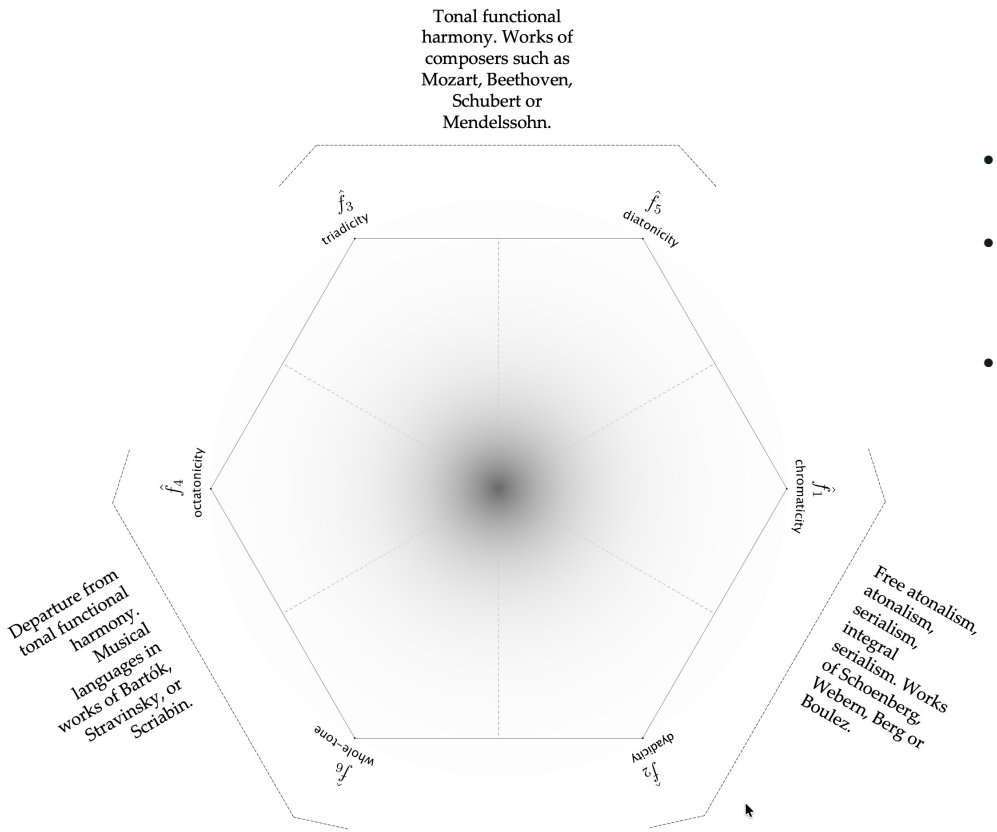


Reading the FQS: Regions and Ambiguity



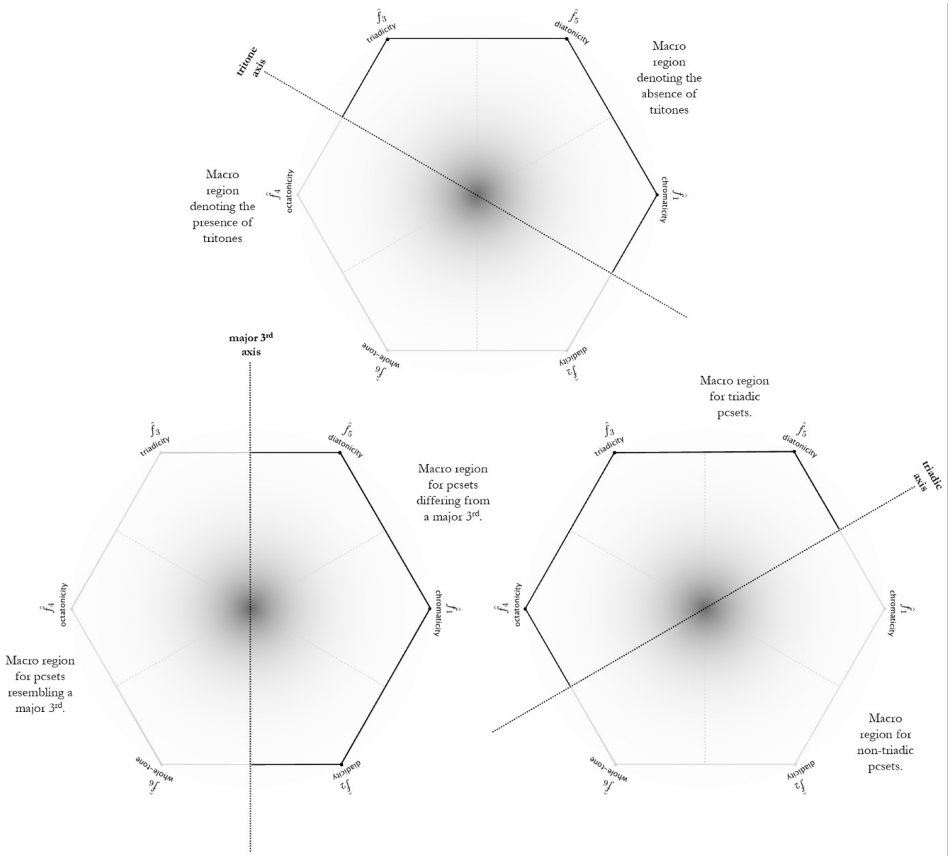
- Six regions aligned with the 6 DFT qualities.
- Centre = "singularity". Point of maximum qualia ambiguity!

Reading the FQS: Regions and Ambiguity



- Six regions aligned with the 6 DFT qualities.
- Centre = “singularity”. Point of maximum qualia ambiguity!
- Three meta-regions:
  - Tonal/functional (triadicity and diatonicity);
  - Musical nationalisms (octatonicity and whole-tone quality);
  - Atonal/serialism (chromaticity and dyadicity).

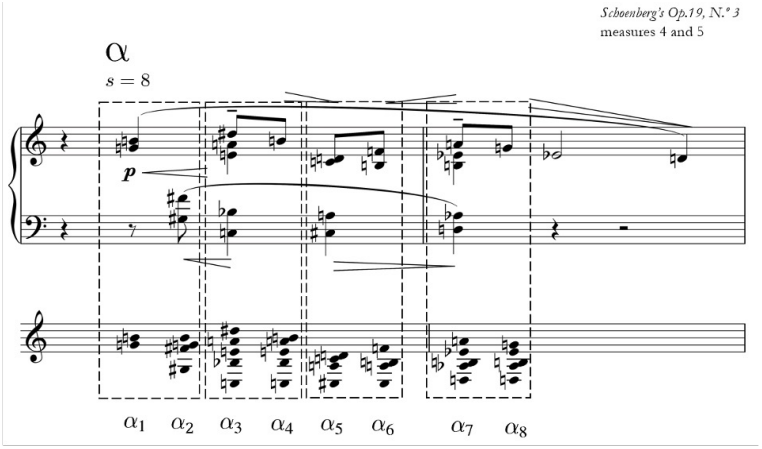
Reading the FQS: Regions and Ambiguity



- Six regions aligned with the 6 DFT qualities.
- Centre = "singularity". Point of maximum qualia ambiguity!
- Three meta-regions:
  - Tonal/functional (triadicity and diatonicity);
  - Musical nationalisms (octatonicity and whole-tone quality);
  - Atonal/serialism (chromaticity and dyadicity).
- Three binary divisons (macro-regions)!

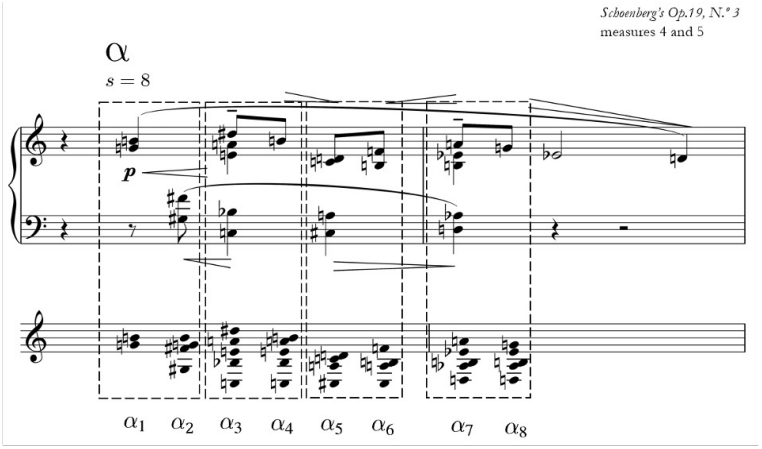
Finding Reference Sonorities

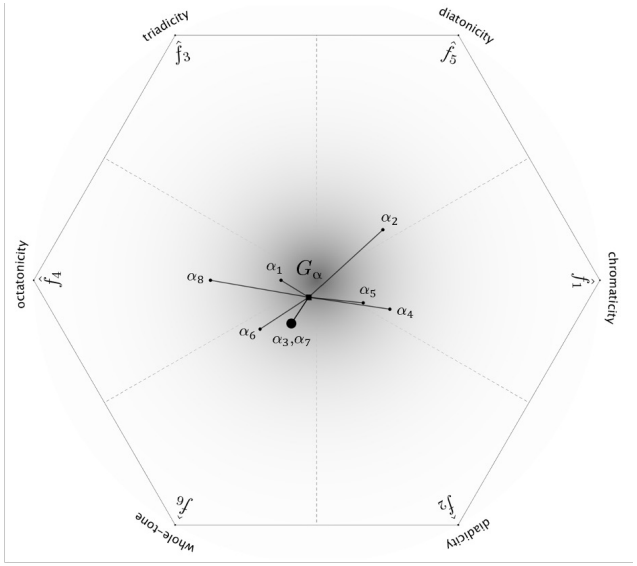
- Method (inspired by Mailman 2015):
    1. Segment pieces into phrases (*phrase-segments*);
    2. Identify each pcset in the segment;



Finding Reference Sonorities

- Method (inspired by Mailman 2015):
    1. Segment pieces into phrases (*phrase-segments*);
    2. Identify each pcset in the segment;
    3. Plot in the FQS each pcset in one segment;
    4. Calculate centroid of all pcsets;
    5. Select nearest actual pcset as reference sonority (RS);
    6. Repeat process for every phrase in a musical piece.

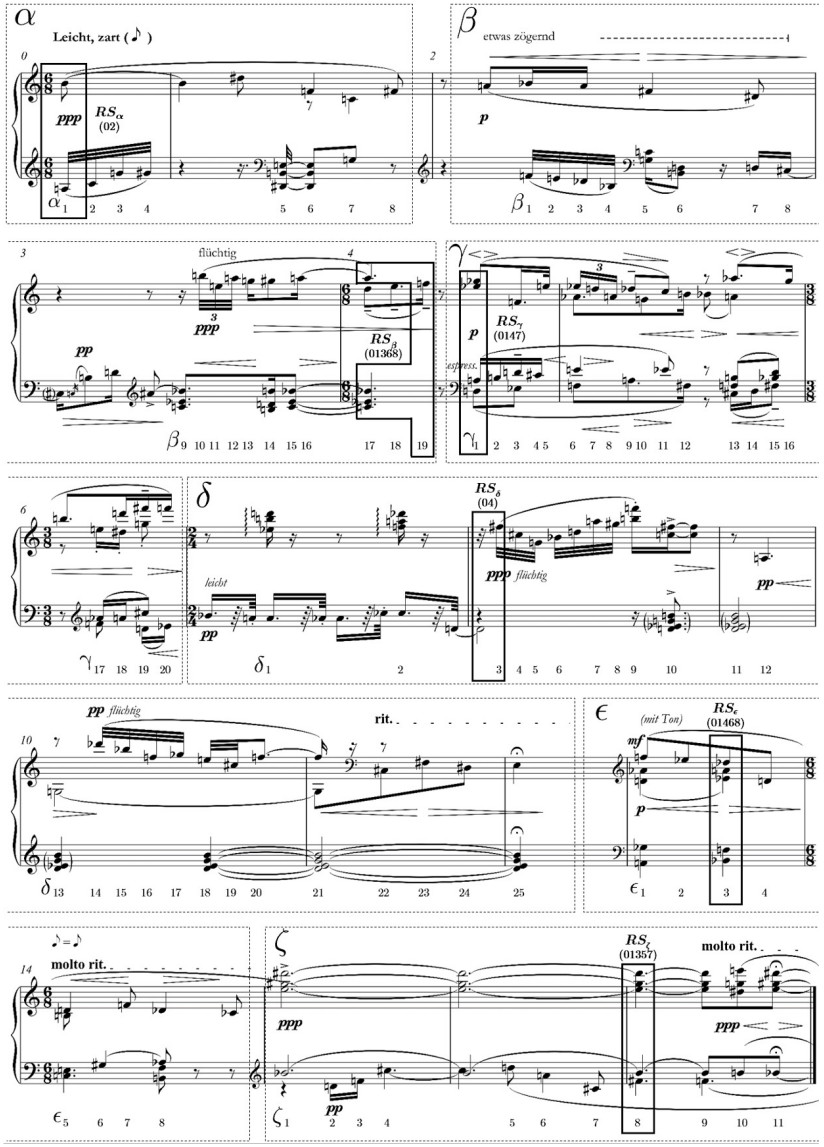




Schoenberg Op. 19, No. 1 (1911)

Phrase	Bars	s	SCs	Aggregated SCs
α	0 and 1	8	$\alpha_1 \mapsto \alpha_8$	$1-8$
β	2 to 4 (1st beat)	19	$\beta_1 \mapsto \beta_{19}$	$9-27$
γ	4 (2nd beat) to 6	20	$\gamma_1 \mapsto \gamma_{20}$	$28-47$
δ	7 to 12	25	$\delta_1 \mapsto \delta_{25}$	$48-72$
ϵ	13 and 14	8	$\epsilon_1 \mapsto \epsilon_8$	$73-80$
ζ	15 to 17	11	$\zeta_1 \mapsto \zeta_{11}$	$81-91$

- Analysis parameters:
  - Six phrases $(\alpha, \beta, \gamma, \delta, \epsilon, \zeta)$;
  - 91 pitch-class sets total;
  - Free atonal period.



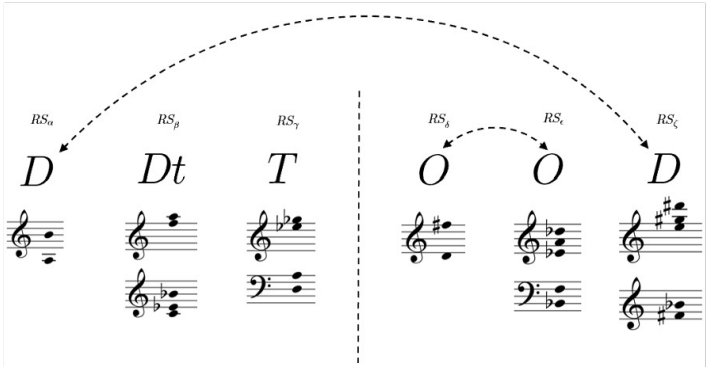
Schoenberg Op. 19, No. 1 (1911)

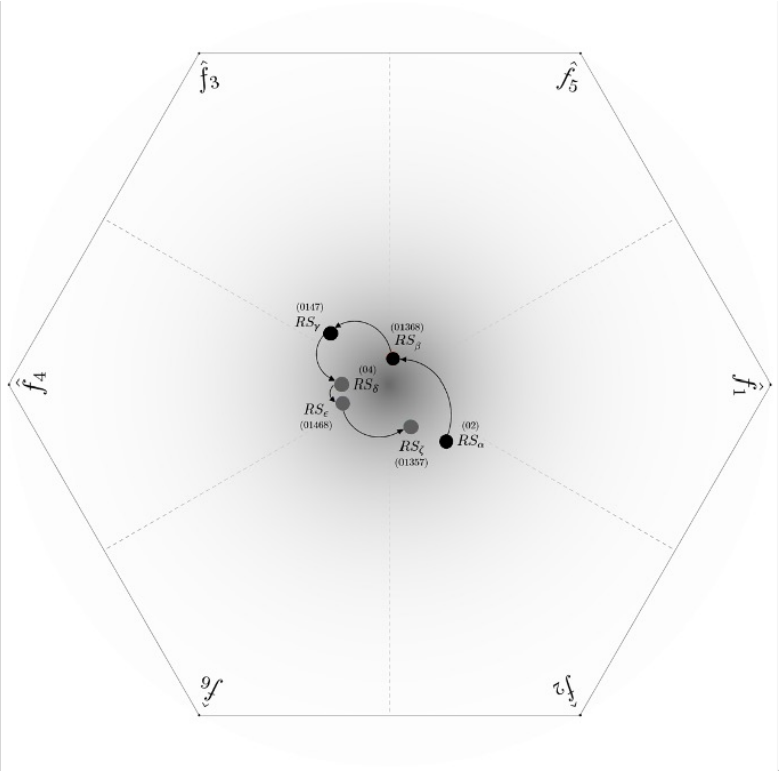
- Analysis parameters:
  - Six phrases ($\alpha, \beta, \gamma, \delta, \epsilon, \zeta$);
  - 91 pitch-class sets total;
  - Free atonal period.

- Question:
  - Can the FQS and the method reveal hidden structural elements?

Discoveries 1 and 2: Circular trajectory and Bipartite Structure

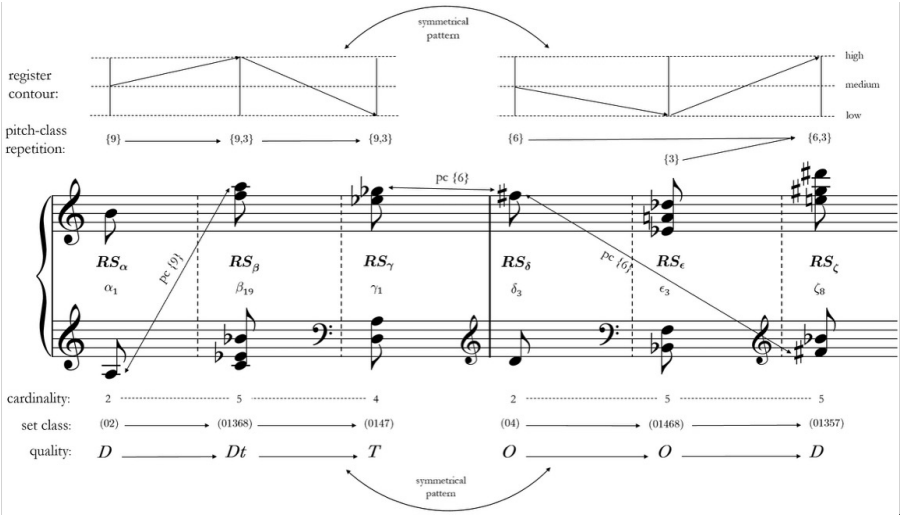
- Anticlockwise circular motion in FQS.
- Dyadicity bookends (absent in middle).



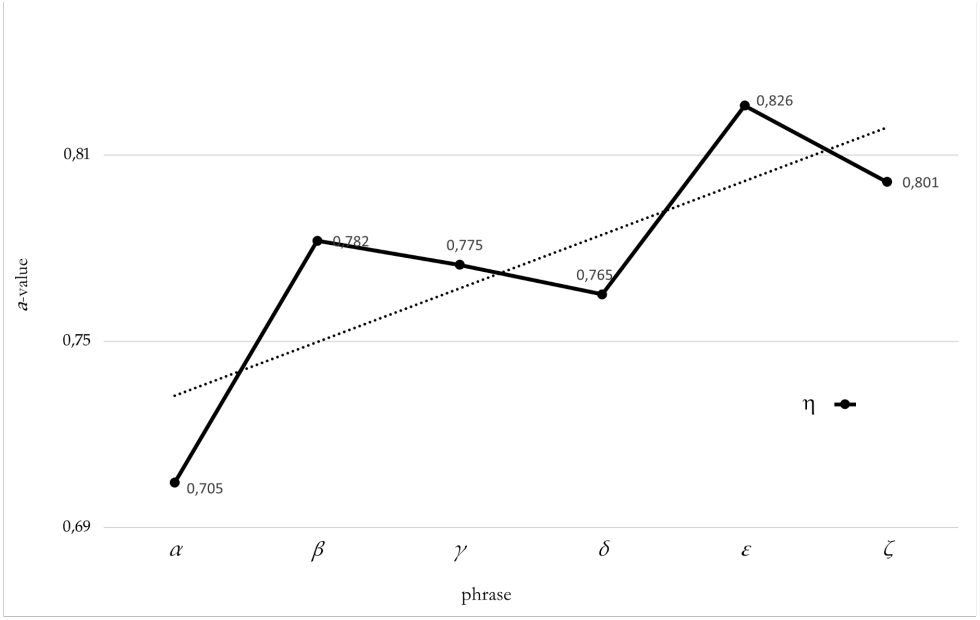


Discoveries 1 and 2: Circular trajectory and Bipartite Structure

- Anticlockwise circular motion in FQS.
- Dyadicity bookends (absent in middle).
- Symmetrical structure: $\alpha \mapsto \beta \mapsto \gamma$ mirrors $\delta \mapsto \epsilon \mapsto \zeta$.
- Set class relationships: $(02) \mapsto (04)$, $(01368) \mapsto (01468)$, and $(0147) \mapsto (01357)$.

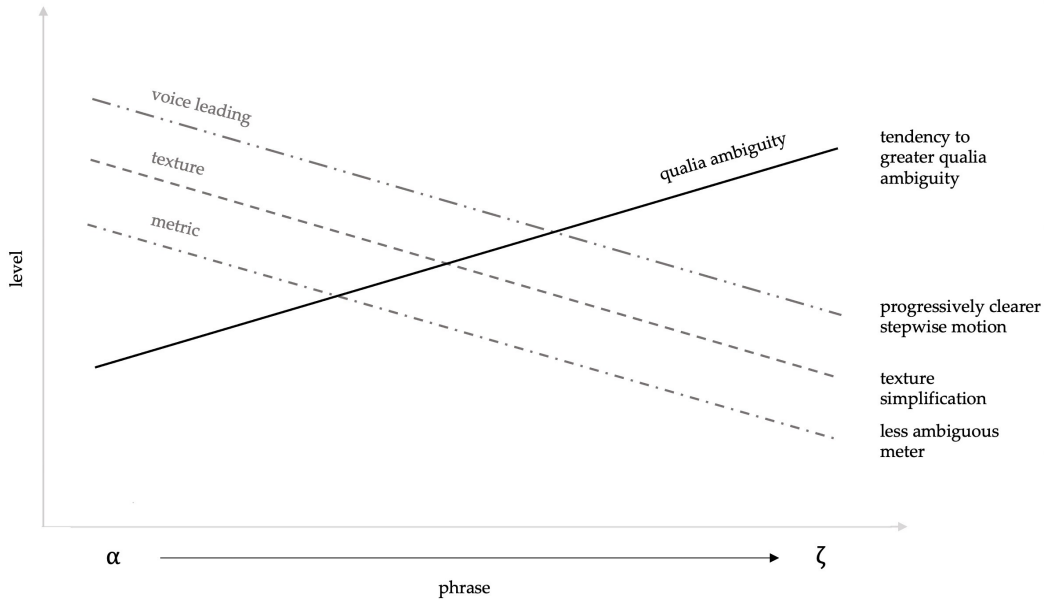


Discovery 3: Linearities



- Kramer's "nondirected linearity".
- Qualia ambiguity increases.

Discovery 3: (Contrapuntal!) Linearities

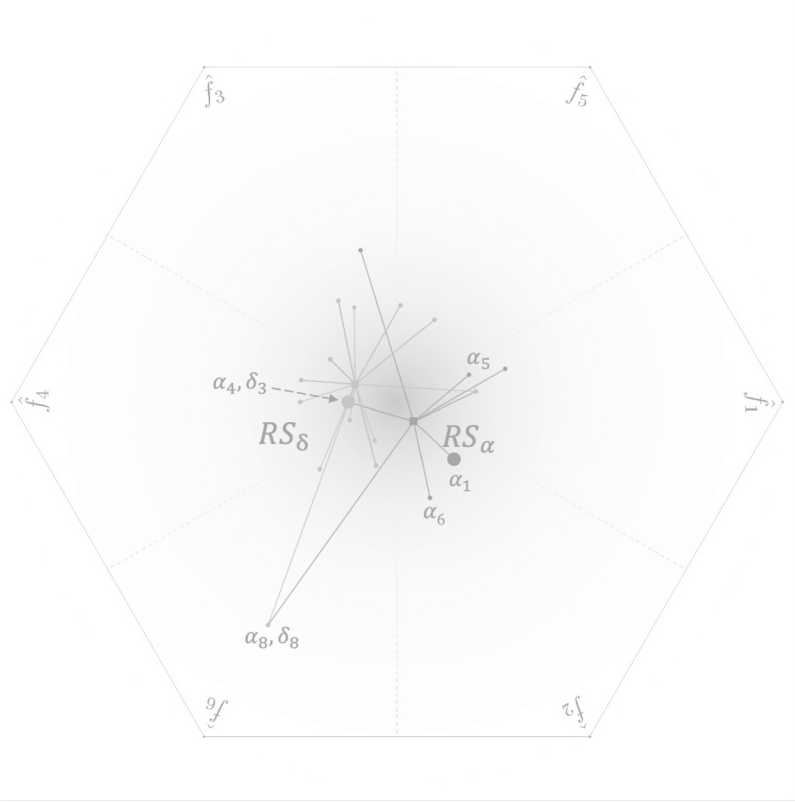


- Kramer's "nondirected linearity".
- Qualia ambiguity increases.
- Contrary motion with other parameters:
  - Voice leading → clearer;
  - Texture → simpler;
  - Metric → more defined;
  - Harmonic Qualia → more ambiguous.

Contributions and Implications

- Main contributions:
  - FQS: full DFT quality space;
  - “Disambiguating ambiguity”: objective metric (η);
  - Revealed hidden symmetries in Schoenberg;
  - Demonstrated contrapuntal linearities.

- Broader implications:
  - Applicable to any atonal/post-tonal repertoire;
  - Bridge between math and musical intuition.



Thank you!